



Quantitative Biophysics and Ecological Systems

BIO 401 Midterm Examination • Time Allowed: 90 Minutes

Student Full Name: _____ Student ID: _____
 Section / Lab: _____ Date: August 28, 2026

Run L^AT_EX again to produce the table

1. Conceptual Biophysics and Bio-Energetics (15 Points)

- (a) (5 points) In enzyme kinetics, how does a competitive reversible inhibitor affect the apparent parameters $V_{\max}$ and K_m ?
- A. $V_{\max}$ decreases; K_m remains unchanged.
 - B. $V_{\max}$ remains unchanged; apparent K_m increases ($K'_m = K_m(1 + [I]/K_i)$).**
 - C. Both $V_{\max}$ and K_m decrease proportionally.
 - D. Both $V_{\max}$ and K_m increase proportionally.

Solution: Correct Choice: B. Competitive inhibitors bind exclusively to the free enzyme active site, requiring higher substrate concentration to reach half-maximal velocity without reducing total enzyme capacity $V_{\max}$.

- (b) (5 points) If primary producers capture 10,000J of energy, approximately how much energy is expected to be incorporated into secondary consumers under Lindeman's 10% law?
- A. 1,000J
 - B. 100J**
 - C. 10J
 - D. 1J

Solution: Correct Choice: B. Tier 1 (Producers) = 10,000J → Tier 2 (Primary Consumers) = 1,000J → Tier 3 (Secondary Consumers) = 100J.

- (c) (5 points) In cooperative allosteric binding modeled by the Hill equation $v = \frac{V_{\max}[S]^h}{K_{0.5}^h + [S]^h}$, a Hill coefficient of $h = 2.8$ indicates:
- A. Hyperbolic non-cooperative kinetics.
 - B. Positive cooperativity with sigmoidal binding kinetics.**
 - C. Negative cooperativity with high substrate inhibition.
 - D. Irreversible enzymatic denaturation.

Solution: Correct Choice: B. $h > 1$ denotes positive cooperativity, where substrate binding at one active site increases binding affinity at adjacent subunits.

2. **Derivation of the Lineweaver-Burk Kinetic Equation (20 Points)** Starting from the Michaelis-Menten rate equation $v = \frac{V_{\max}[S]}{K_m + [S]}$:

(a) (10 points) Take the algebraic reciprocal of both sides to derive the double-reciprocal (Lineweaver-Burk) linear transformation:

$$\frac{1}{v} = \frac{K_m}{V_{\max}} \frac{1}{[S]} + \frac{1}{V_{\max}}$$

Solution:

$$\frac{1}{v} = \frac{K_m + [S]}{V_{\max}[S]} = \frac{K_m}{V_{\max}[S]} + \frac{[S]}{V_{\max}[S]} = \left(\frac{K_m}{V_{\max}} \right) \frac{1}{[S]} + \frac{1}{V_{\max}} \quad \blacksquare$$

(b) (10 points) Identify the y-intercept, slope, and x-intercept of the resulting plot of $1/v$ against $1/[S]$.

Solution:

- **y-intercept** ($1/[S] = 0$): $\frac{1}{V_{\max}}$
- **Slope** (m): $\frac{K_m}{V_{\max}}$
- **x-intercept** ($1/v = 0$): $-\frac{1}{K_m}$